ELECTRIFICATION STATUS OF HEALTH FACILITIES

This is a short survey, to be completed only by the facility's Medical Director, Head Nurse, or Administrator/Manager. Some questions are technical and may require assistance from a technician.

We are conducting a national survey to better understand the electrification status of all health facilities.

This survey is led by the Humanitarian Aid Research Center (RCHA-RDC asbl) and the Renewable and Appropriate Energy Laboratory at UC Berkeley (RAEL), in partnership with the Provincial Health Divisions of North Kivu, South Kivu, and Haut Katanga.

Section: General Information

- Date: _____
- Email: _____
- Name: _____
- Phone Number: _____
- Your Role/Position in the Facility (select one):
    - Medical Director
    - Doctor
    - Head Nurse
    - Administrator / Manager
    - Technician

Facility Details

- Province (select one):
    - North Kivu
    - South Kivu
    - Haut Katanga
- Health Zone (select one): _____
- Type of Facility (select one):
    - HGR (General Reference Hospital)
    - CH (Hospital Center)
    - CSR (Reference Health Center)
    - HP (Provincial Hospital)
    - CM (Medical Center)
- Facility Name (select one): _____

- What is the approximate population served by your health facility (catchment area population)?: _____

Energy Sources

- Does your facility have an Internet connection? (select one):
    - No
    - Yes

Energy Sources – Grid

- Is your facility connected to an electrical grid? (select one):
    - SNEL
    - SOCODE
    - VIRUNGA SARL ("De Merode")
    - NURU SARL
    - ENK (Energies du Nord Kivu)
    - Equatorial Power
    - Community Network – Hydro
    - Community Network – Solar
    - No grid
    - Other grid
- Which other grid?: _____
- When was the facility connected to the grid?: _____
- Average monthly cost: _____

Other Energy Sources – Generators

- Does your facility have a generator? (select one):
    - No
    - Yes
- How many generators does your facility have?: _____
- Capacity of first generator (kVA): _____
- When was it acquired?: _____
- Does the generator start automatically when power goes out? (select one):
    - No
    - Yes

(repeat for second and third generator as above)

- Average monthly cost for all generators: _____
- Frequency of use (select one):
    - More than 10 hours per day
    - Every day, more than 1 hour at a time
    - Every day, less than 1 hour at a time
    - Not every day, but several times per week
    - A few times per month
- What level of quality do generators provide? (select one):
    - Power all equipment in the facility
    - Can only power certain blocks
    - Cannot power very large equipment
    - Only provide lighting and phone charging

Other Energy Sources – Solar System

- Does your facility have a solar system? (select one):
    - No
    - Yes
- When was the solar system installed at the facility?: _____
- Number of solar panels installed: _____
- Number of batteries installed: _____
- Age of current batteries (years): _____
- Total system capacity (kWp): _____
- What level of quality does the solar system provide? (select one):
    - Powers all equipment in the facility
    - Can only power certain blocks
    - Cannot power very large equipment
    - Only provides lighting and phone charging
    - The system no longer works at all
- How is the solar system financed and maintained? (select one):
    - Fully donated; the facility pays nothing (no upfront cost, no monthly fee)
    - Donated, but the facility pays for maintenance when needed
    - Donated, but the facility pays a monthly fee for maintenance
    - Facility paid full upfront cost and pays for maintenance when needed
    - Facility paid full upfront cost and pays a monthly fee for maintenance
    - Facility pays a monthly fee through a financing or leasing plan and pays for maintenance when needed
    - Facility pays a monthly fee through a financing or leasing plan and also pays a monthly fee for maintenance

    - Other, specify
- Total value of all equipment associated with the solar installation: _____
- Average monthly cost paid for the solar system itself (excluding maintenance): _____
- Average monthly cost paid specifically for solar system maintenance: _____
- Typical cost paid for maintenance or repair when required: _____
- Name of the program or donor who provided the solar system: _____

Other Energy Sources – Charcoal

- Does your facility use charcoal? (select one):
    - No
    - Yes
- For what purposes does your facility use charcoal? (select one):
    - Preparing meals for patients or staff
    - Boiling water for hygiene or patients
    - Sterilizing medical instruments
    - Heating water for postnatal care or newborn baths
    - Other, specify
- Other, specify: _____
- Average monthly cost: _____

Outages

We want to better understand the reliability of electricity in your facility. We know outages are not always constant—some days are better, others worse. We aim to capture a clear picture of both typical and extreme cases.

- In the past 7 days, was electricity (excluding backup generator) available during operating hours, or were there interruptions longer than 2 consecutive hours? (select one):
    - Always available (no interruptions)
    - Often available (interruptions less than 2 hours per day)
    - Sometimes available (frequent or prolonged interruptions > 2 hours per day)
- During a typical week, how many days is electricity continuously available (24 hours) without any outages?: _____

- On a typical day, how many total hours is electricity unavailable?: _____
- How many days per week are “typical” in terms of electricity availability?: _____
- On a “bad day,” how many total hours is electricity unavailable?: _____
- How often does your facility experience a “bad day”? (select one):
  - Less than once per month
  - 1–3 days per month
  - Once per week
  - 2–3 times per week
  - Other, specify
- Other, specify: _____
- Does your facility experience very short outages (less than 5 minutes)? (select one):
  - No
  - Yes
- If yes, how many times do these short outages occur on a typical day?: _____

Recall – Last Three Days

Please estimate the number of outages, the total outage time, and whether the day was good, typical, or bad for each of the past three days.

- Yesterday – Number of outages: _____
- Yesterday – Total outage time (hours): _____
- Yesterday – Type of day (select one): Good / Typical / Bad

(repeat for 2 days ago, 3 days ago)

Maintenance of Energy Sources & Medical Equipment

- How are all energy sources managed and maintained at your facility? (select one):
  - We call the energy provider (e.g., grid operator) when there is a problem
  - We have an on-site technician who manages energy systems
  - We call a local community technician as needed
  - There is no maintenance plan for energy systems
  - Other, specify
- Other, specify: _____
- How is electrical wiring installed at your facility? (select one):

    - Single-phase (one power line)
    - Three-phase (three power lines)
    - Don't know
- Have any of your medical devices been damaged due to poor electricity supply? (select one):
    - No
    - Yes
- If yes, which ones? Provide details on non-functional equipment and causes: _____
- Number of 10L oxygen concentrators in your facility: _____
- Number of 5L oxygen concentrators in your facility: _____
- How else does your facility obtain oxygen (other than concentrators)? (select one):
    - Oxygen cylinders
    - On-site PSA plant
    - External PSA supplier
    - No oxygen available
    - Other, specify
- Other, specify: _____
- Do patients have to pay for oxygen? (select one):
    - Yes, included in general treatment costs
    - Yes, paid separately
    - No, provided free of charge
- Cost of oxygen: _____

Patient Records and Health Data

- Do you keep a patient register? (select one):
    - Yes, only a central register
    - Yes, each service has its own register
    - Yes, both service registers and a central register
    - Other, specify
- Other, specify: _____
- Which services have their own registers? (select one):
    - Surgery
    - Maternity
    - Pediatrics
    - Inpatient/Internal medicine
    - Outpatient/Consultation
    - Emergency
    - Other, specify

- Other, specify: _____

- Do you keep a register of deaths? (select one):
    - No
    - Yes
- How do you record deaths in your facility?: _____

- Do you have a morgue? (select one):
    - No
    - Yes
- Do you keep a register for the morgue? (select one):
    - No
    - Yes

Final Questions

- Is electricity theft a problem in the area around your facility? (select one):
    - No, not a problem
    - Yes, a minor problem
    - Yes, a major problem
- If we install at our cost a 2.7 kW solar system with 5 kWh lithium-ion storage, able to power equipment such as an oxygen concentrator, water pump, or autoclave 24/7, with 10 years guaranteed maintenance: how much would your facility be willing to pay per month (USD)?: _____

- Is Internet available in your facility? (select one):
    - No
    - Yes
- Did you request help from a technician or staff member to complete this survey? (select one):
    - No
    - Yes
- What other problems does your facility face that our consortium could help address?: _____

- Is there anything else we should know about your facility?: _____